Shared Load, No Theater: Active/Active HAProxy Load Balancing with Dual VIPs and Keepalived

Emphasizes real concurrent traffic handling—not just standby failover.

Alright, fresh Ubuntu VMs on bridge connections - **perfect**. No more WiFi nonsense. Here's the **complete, copy-paste deployment** for your two dedicated load balancers.

🎯 Your Network Map

```
              +-----------------------+
              |   Wazuh Agents/API    |
              +-----------+-----------+
                          |
        +-----------------+-----------------+
        |                                   |
   VIP1: 192.168.0.100               VIP2: 192.168.0.101
   (MASTER on balancer2)               (MASTER on balancer3)
        |                                   |
   +------------------------------+     +------------------------------+
   |  balancer2 (Ubuntu 22.04)    |     | balancer3 (Ubuntu 22.04)     |
   |  ens33: 192.168.0.108        |     | ens33: 192.168.0.105         |
   +------------------------------+     +------------------------------+
                  |                               |
                  +-----------+-----------+-------------+
```

```
                    │         │
            Wazuh Manager: 192.168.0.104 (all ports)
```

Step 1: Installation Script (Run on BOTH nodes)

```
#!/bin/bash
# Install HAProxy + Keepalived on Ubuntu 22.04

# Update system
sudo apt update && sudo apt upgrade -y

# Install packages
sudo apt install -y haproxy keepalived socat

# Enable HAProxy (Ubuntu 22.04 needs manual enable)
sudo systemctl enable haproxy
sudo systemctl enable keepalived

# Backup default configs
sudo cp /etc/haproxy/haproxy.cfg{,.bak}
sudo cp /etc/keepalived/keepalived.conf{,.bak}

echo "Installation complete. Proceed to configuration."
```

Save as `install.sh`, make executable: `chmod +x install.sh && ./install.sh`

Step 2: Keepalived Config for balancer2 (192.168.0.108)

```
sudo tee /etc/keepalived/keepalived.conf <<'EOF'
vrrp_instance VI_1 {
    state MASTER
    interface ens33
```

```
    virtual_router_id 51
    priority 150
    advert_int 1

    # Unicast for bridge network stability
    unicast_src_ip 192.168.0.108
    unicast_peer { 192.168.0.105 }

    authentication {
        auth_type PASS
        auth_pass WazuhHA2024!
    }

    virtual_ipaddress {
        192.168.0.100/24 dev ens33 label ens33:vip1
    }
}

vrrp_instance VI_2 {
    state BACKUP
    interface ens33
    virtual_router_id 52
    priority 100
    advert_int 1

    unicast_src_ip 192.168.0.108
    unicast_peer { 192.168.0.105 }

    authentication {
        auth_type PASS
        auth_pass WazuhHA2024!
    }

    virtual_ipaddress {
        192.168.0.101/24 dev ens33 label ens33:vip2
    }
```

```
}
EOF
```

Step 3: Keepalived Config for balancer3 (192.168.0.105)

```
sudo tee /etc/keepalived/keepalived.conf <<'EOF'
vrrp_instance VI_1 {
   state BACKUP
   interface ens33
   virtual_router_id 51
   priority 100
   advert_int 1

   unicast_src_ip 192.168.0.105
   unicast_peer { 192.168.0.108 }

   authentication {
      auth_type PASS
      auth_pass WazuhHA2024!
   }

   virtual_ipaddress {
      192.168.0.100/24 dev ens33 label ens33:vip1
   }
}

vrrp_instance VI_2 {
   state MASTER
   interface ens33
   virtual_router_id 52
   priority 150
   advert_int 1
```

```
    unicast_src_ip 192.168.0.105
    unicast_peer { 192.168.0.108 }

    authentication {
        auth_type PASS
        auth_pass WazuhHA2024!
    }

    virtual_ipaddress {
        192.168.0.101/24 dev ens33 label ens33:vip2
    }
}
EOF
```

Step 4: HAProxy Config (SAME on BOTH nodes)

```
sudo tee /etc/haproxy/haproxy.cfg <<'EOF'
global
    log /dev/log local0 info
    chroot /var/lib/haproxy
    stats socket /run/haproxy/admin.sock mode 660 level admin expose-fd liste
ners
    stats timeout 30s
    user haproxy
    group haproxy
    daemon

defaults
    log global
    mode tcp
    option tcplog
    timeout connect 5s
    timeout client 60s
    timeout server 60s
```

```
    timeout check 2s

# Wazuh Agent Events (1514)
frontend wazuh_agent
    bind *:1514
    mode tcp
    default_backend wazuh_managers

backend wazuh_managers
    mode tcp
    balance roundrobin
    option tcp-check
    server wazuh-mgr-1 192.168.0.104:1514 check inter 5s rise 2 fall 3

# Wazuh Registration (1515)
frontend wazuh_registration
    bind *:1515
    mode tcp
    default_backend wazuh_registration_backend

backend wazuh_registration_backend
    mode tcp
    balance roundrobin
    option tcp-check
    server wazuh-mgr-1 192.168.0.104:1515 check inter 5s rise 2 fall 3

# Wazuh API (55000)
frontend wazuh_api
    bind *:55000
    mode tcp
    default_backend wazuh_api_backend

backend wazuh_api_backend
    mode tcp
    balance roundrobin
    option tcp-check
```

```
    server wazuh-mgr-1 192.168.0.104:55000 check inter 5s rise 2 fall 3

# HAProxy Stats (8404)
frontend stats
    bind *:8404
    stats enable
    stats uri /
    stats refresh 10s
    stats admin if TRUE
EOF
```

Step 5: Firewall Configuration (BOTH nodes)

```
# Reset ufw and configure
sudo ufw --force reset
sudo ufw default deny incoming
sudo ufw default allow outgoing

# Allow HAProxy ports
sudo ufw allow 1514,1515,55000,8404/tcp

# Allow VRRP (Keepalived)
sudo ufw allow proto vrrp from 192.168.0.0/24

# Allow SSH
sudo ufw allow 22/tcp

# Enable firewall
sudo ufw --force enable
sudo ufw status verbose
```

Step 6: Start Services (BOTH nodes)

```
# Restart services in correct order
sudo systemctl daemon-reload
sudo systemctl restart keepalived
sudo systemctl restart haproxy

# Check status
sudo systemctl status keepalived haproxy --no-pager

# Enable auto-start
sudo systemctl enable keepalived haproxy
```

Step 7: Verification Commands

Check VIPs

```
# On balancer2 (should have 192.168.0.100)
ip a show ens33 | grep "inet "

# On balancer3 (should have 192.168.0.101)
ip a show ens33 | grep "inet "
```

Test Failover

```
# On balancer2, trigger failover
sudo systemctl stop keepalived

# On balancer3, verify it took both VIPs
ip a show ens33 | grep "192.168.0.100"
```

Check HAProxy Backend Health

```
# View real-time stats
echo "show stat" | sudo socat /run/haproxy/admin.sock stdio
```

```
# Check specific backend
echo "show stat" | sudo socat /run/haproxy/admin.sock stdio | grep wazuh-mgr-1
```

Access Stats Page

```
<http://192.168.0.100:8404>
<http://192.168.0.101:8404>
```

Step 8: Ubuntu 22.04 Specific Troubleshooting

Issue	Fix
HAProxy fails to bind	Check `netstat -tlnp | grep 1514` - stop any Wazuh services on LB nodes
Keepalived multicast warnings	Ignore - unicast is active
VIP not moving on failover	Run `sudo killall -USR1 keepalived` to force VRRP sync
Bridge network latency	Add `vrrp_garp_master_delay 1` to both configs
Permission denied on stats socket	Add user to haproxy group: `sudo usermod -aG haproxy $USER`

Check VRRP communication:

```
# From balancer2
sudo tcpdump -i ens33 -n vrrp

# Should show advertisements between 192.168.0.108 and 192.168.0.105
```

Step 9: Agent Configuration

On your Wazuh agents, use **either VIP**:

```xml
<server>
  <address>192.168.0.100</address>
  <port>1514</port>
  <protocol>tcp</protocol>
</server>
```

Or for load distribution:

```
# Half your agents use 192.168.0.100, half use 192.168.0.101
```

Done. Run Steps 1-6 on both VMs. Your Active-Active HAProxy cluster is ready.

Need next? Options: SSL termination, multi-manager backend, monitoring, or agent enrollment automation.